ENGINE SYSTEMS

Fuel System

Component Technical Specification (CTS)

for

SMART FUEL FILTER

S K0

REVISION TABLE

RELEASE	DATA	UPDATE

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1. INTRODUCTION

1.1. Scope of document

This specification establishes the Smart Fuel Filter requirements for the S K0 application. The function, performance, reliability, interface and validation requirements are defined in this document. They are applicable for the specified markets and associated ambient environments here detailed.

1.2. System description

The Filters are designed to provide a proper flow rate and pressure to the engine. Additionally, Filters have to guarantee an adequate filtration level and have to indicate the fuel level inside the tank.
The component is available in the following configurations that are listed in Table 1.2.

Table 1.2- Powertrain Configurations

Engine identification	Displacement [dm^3]	FUEL	Market	Ecology
FamB 2.2L Mjet	2.2	DIESEL	Enlarged Europe & LATAM	€7 / PL8

The subsystem is available to supplier in the following configurations that are shown in the hereafter attached file:

02b_CTS att_ K0
_ SFF _2025_11_0

For all dimensions refer to 3D model.

The previous preliminary Filter CAD Data has to be considered a preliminary path. They will be changed later on according to car development. RASI chart states rules for this further development. CAD drawing standard has to be fully compliant with requirements of §2.3.5. CAD drawing standard has to be fully compliant with guidelines for using the latest version of UG/Teamcenter or CATIA used by S in Design applications. All changes to the CAD model allowed to suppliers regarding Filter geometry have to be fully compliant with S Package Specification PRO.00004 and Design Guideline PRO.00004.
Supplier has to be updated and has to share S Package Team Requirements through all phases of component development.
All parts of Filter have to be designed by Supplier properly in order to guarantee proper functionality of Filter assembly.
Alternative solution shall guarantee, if compared to mainstream solution, a lower weight, a lower cost, an easier assembly or a combination of the previous aspects. All solutions in terms of divisions of the hose or materials are accepted fulfilling anyway all technical requirements listed in the present specification. Information contained in §1.2 shall be considered as a target for mainstream solution: every deviation from this target shall be clearly explained in the “Deviation List” and during the Technical Review.

1.3. Packaging protection

NOT APPLICABLE

2. APPLICABLE DOCUMENTS

2.1. Order of precedence

In the event of a conflict between the text of this specification and the documents cited herein, the text of this specification takes precedence. In a conflict between requirements of S specifications, the specification is valid, which has the higher requirements. Nothing in this specification, however, supersedes applicable government laws and regulations unless a specific exemption has been obtained.

2.2. Government documents

- EC (European Community)
- ECE (Economic Commission Europe)

2.3. Drawing standards

Following an example of the technical information that supplier will need to include inside the drawing of the component:

Note: this is not the drawing of this applications.

2.4. documents

The following documents have to be considered for development of the subsystem.

NOTE: Engineering team is not allowed to send by mail specifications.
All the norms are available on the web portal " eSupplierConnect.com ".

2.4.1. General Requirements for Suppliers

	S Specification
Supplier Quality	9.01102
General Purchasing conditions	9.01100
Product quality and conformity certificate	9.01103
Qualification procedure	07740
Product FMEA	CS.00133
Process FMEA	00271
Supplier code	07610
Production date	0.00013
Procedure for the individuation of integration level of product [Supplier integration level 2]	08034
Trademark identification on parts	CS.11000
Material Preservation	71068

2.4.2. Component specification

	S Spec.
DIESEL FILTERS FOR INJECTION ENGINES - 01446_24_00041	PF.90150
S ERGONOMICS GUIDELINES - DMOV_CAR02_0017	CEP.00082
QUICK CONNECT - FUEL SYSTEM	PF.90298
Miscellaneous plastic component (Thermoplastic or thermosetting)	9.55253/MS
O-Ring	14618
PLASTICS - POLYACETAL 01446_24_00035	MS.50210
S ELECTRICAL AND EMC PERFORMANCE REQUIREMENTS FOR EE COMPONENTS - 01446_23_00323	CS.00244
ENVIRONMENTAL SPECIFICATION FOR ELECTRICAL / ELECTRONIC (E/E) COMPONENTS - 01446_24_00439	CS.00263

2.4.3. Material Requirements

	S Specification
SUBSTANCES OF CONCERN REQUIREMENTS FOR VEHICLE, PARTS AND MATERIALS - 01446_24_00537	CS.00265
Recycling markings for plastic, composite and elastomer components	07416

Recyclability and recoverability of a vehicle	00256

Environmental Standard

Supplier will provide the list of all materials according to the guideline released by S.

Alternative materials will be evaluated by S during the sourcing process.

Main guidelines:

- The supplier is required to avoid to use PVC wherever is possible, using alternative materials which not affecting the performances/cost of the system. In case the alternative solution would lead to a cost increase, the supplier will propose a technical/economical solution which will be evaluated by S.
- All the materials must be consistent to S 9.01102 annex CK. The materials used in the Filter should be recyclable as much as possible after the vehicle disassembly. Maximum allowable regrind is 10% per S Materials Engineering Team.*
- Marking on the Filter must be consistent to S 07416. Calculate the recyclability according to S 00256 (green list). Point out eventual criticalities to Material dept.
- The supplier must use the IMDS system for the materials recyclability. The production approval will be given only after S has received the material data sheet via IMDS system.

*NOTE: Deviations shall be agreed with S R&D team and clearly explained in the “Deviation List” and during the Technical Review.

2.4.1. Addendum

	S Specification
Global gage standard for plastic components	ME.TAG002
Manufacturing Requirements for Supplier (MRS) 01598_16_04265	GSQN.010
Quality assessment of painting of supplied components according to custumer perception	FGP.28

Anodizing treatment for aluminium and aluminium alloy parts - decorative and corrosion resistance	PS.50050
Supplier evaluation for superficial and/or aesthetical treatments	FGP.34
General requirements for released drawings	CS.00022
Product identification label	9.01105
Material preservation	71068

2.4.2. Design Requirements

	S Specification
Procedure for vehicle part marking MOPAR LOGO, “F solution”;	CS.11000
CAD drawings, CAD modelling/drawings rules	CS.00080
TcAE system and CAD part management within the scope of product/process development	07226
CAD drawings - Rules for 3DA documentation definition (3D Annotated Models)	CS.00029
CAD drawings – Self-certification of the quality of the CAD parts (check list)	07247
CAD drawings – Self-certification of graphs quality (checklist) for E&D (apart from wiring harnesses)	07247/01
New drawing title block – definitions, completion rules and support tables	CS.00003
Dimensioning and tolerancing	ASME Y14.5-2009
Geometric dimensioning and tolerancing practices (ASME addendum)	CS.00019

3. PERFORMANCE/FUNCTIONAL REQUIREMENT

3.1. Functional Content

The component has to provide the following functions:

a. guarantee working condition under the following boundary conditions:
   - Ambient environment
     - Working ambient temperature +45°C to 120°C
     - Working Fuel temperature -45°C to 90°C
     - Operating Fuel pressure 450 to 700 kPa
     - Relative humidity in use: 0 to 100% / Storage: max. 60%
     - For extreme transport condition considers storage temperature has to be between -30°C and +70°C. The atmospheric humidity must not exceed 60 % during storage for 6 months.

   - The performances must be guaranteed also with -3 sigma pump and the filter at end of life and the all electrical components degradable at end of life

b. Filter main characteristics must be as following:
   - Nominal Flow Rate 190 l/h

   - Pressure drop according to ISO 4020 :
     10kPa

   - Initial Filtration Efficiency according to PF.90150 section7.1 (ISO 19438):
     98% @4 um @ Nominal Flow Rate

   - Dust Holding Capacity according to PF.90150 section 7.3 (ISO 4020):
     30g @ 500kPa @ end of life pressure

   - Water Separation efficiency according to PF.90150 section 7.12 (ISO 16332):
     93% @ Nominal Flow Rate

   - Water storage capacity :
     150 ml

   - Waler alarm level :
     100 ml

3.2. Hydraulic interfaces

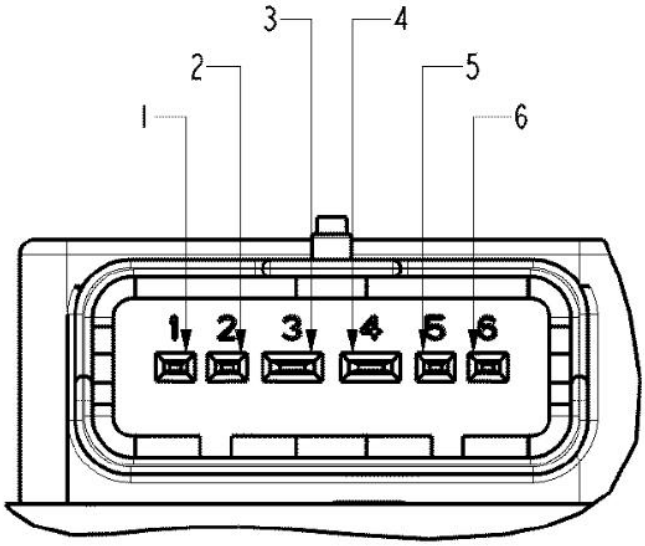
Each interface spigot shall be according to CAD data and S Std. PF.90298 / SAE J2044.

Each interface should be in according to the following table and will be shared with component specialist during D-FMEA.

Connection	Type	Reference spec
Inlet port	10mm diameter	B12 2813
Outlet port	10mm diameter	B12 2813

3.3. Electrical interfaces

Electrical connector need to be according to APTIV p/n 211FYT0523 , and using the following pinout:



CONNECTOR INTERFACE DESIGN
ACCORDING TO APTIV CONNECTOR
P/N:211FT0523 - KEY CODE A
DETAIL B (2:1)

PIN	FUNCTION
1	+5V Power Supply
2	WiF Digital Output
3	PTC -
4	PTC +
5	SENT Protocol
6	GROUND (Sensors)

3.4. Component protection

Protection caps are requested on all port extremities and on electric connector. Protection caps could be eliminated only with specific agreement with Plant UTE and S Quality dept.
Transport conditions of the new type of component storage must be shared with Manufacturing Engineering dept. Protection caps have to be shaped according to following criteria:

protection cap.xlsx

4. VALIDATION

For the validation the supplier has to provide during the offer DV plans for Filter and sub-components and PV (AQF) plan for Filter and sub- components, according the format of the norm 07740 annex 2.

DV PLAN FORMAT.xlsx

PV PLAN FORMAT.xlsx

Below you can find some guidelines for the correct writing of the plan:

- The list of the tests must be done according the S specifications PF.90150, and all tests must be listed, even if are not planned. In case of extension from other equivalent specification or different test procedure you have to write your proposal on the NOTE column.
- If you don't plan to perform some the tests on DV (due to the part not representative or some other cause) write it on the NOTE column and color the line in grey.
- In the same way for PV, if you extend any test from the DV (for example because the part is already OTOP) write it on the NOTE column and color the line in grey.
- On PV plan is mandatory to fill column of requalification tests according to S SPC 07740
- It is required to supplier to provide DV and PV plan for complement components (i.e. ICV ecc.)

Due to the short timing development, it is recommended to the supplier to reuse as much as possible component already validated or with validation on going for other S applications.

5. ADDITIONAL REQUIREMENTS

5.1. Communality strategies

To minimize the costs, as much common components as possible should be used.

5.2. Manufacturability and Assembly

The subsystem shall fulfil the following manufacturing and assembly requirements.

ME Assembly

5.2.1. Serviceability

The following paragraphs define the constraints and requirements associated with serviceability. Serviceability is the ease and cost of performing the maintenance, diagnosis, and repair operations necessary to preserve the original performance level of the vehicle throughout its useful life.
This section contains serviceability requirements.

5.2.1.1. General Requirements

Shall be replaceable as a module. Production parts shall be identical to service parts and vice versa.

5.2.1.2. Damage or Degradation Due To Service

Service, Maintenance and Repair procedures shall be completed without damage or degradation to the serviced part, or related parts/components.

5.2.1.3. Risk of Injury

The correct performance of service, maintenance and repair tasks shall not subject the individual performing the task to unreasonable risk of injury.

5.2.1.4. Service Tools

5.2.1.4.1. Standard Tools

Standard tools shall be used to ensure reasonable repair costs.
Example: EXTERNAL TORX or HEX-HEAD shall be used

5.2.1.4.2. Special Tools

No new special tools shall be required for service or repair

5.2.1.4.3. Service Equipment

No special service equipment should be required.

5.2.1.5. Maintenance / Repair

The following paragraphs define the constraints associated with maintenance/repair requirements and procedures

5.2.1.5.1. Performance of Repair

For performance of repair no tools shall be required.
Replacement task shall not require the removal of unrelated components:

5.2.1.5.2. Maintenance Schedule

Shall be maintenance free (definition of maintenance free ≥ 240.000 km)

5.2.1.6. Service Parts Engineering

5.2.1.6.1. Storage of Parts

Storage of parts will be directed by the part cost and necessity of repair. The design and purchase of system/components must take appropriate parts levels into consideration.

5.2.1.6.2. Part Preparation

Variants of parts must be avoided. Various suppliers or design modifications shall not put obstacles in the way of using the same or modified part even when repairing an older car. If this is not possible, the production tool under no circumstances may be changed. Drawings and specifications shall always be with the right part number, and shall be updated when modifications affect parts.

5.2.1.7. Repair

5.2.1.7.1. Subsystem Repair

The subsystem shall meet the following requirements associated with repair.

5.2.1.7.2. General Requirements

a. Repair procedures shall be defined during development and appropriate information including technical data be released for service readiness operations.
b. Before Tooling Kick Off, detailed service related system descriptions and technical data shall be available.
c. V.P. vehicles shall have 100% okay parts for production of Service Literature.
d. Simple repair methods of subsystem components shall be defined.
e. Subsystem parts shall be available individually to allow economical repairs. The splitting of the item into individual parts has to be agreed between S R&D and Aftersales Engineering.
f. Individual parts of a subsystem assembly shall be marked with their own part numbers.
g. Attachments shall be able to remove/refit at least 10 times without damages to the basis, or its function deteriorate, even when self-drilling fasteners are used.
h. Hex-head or torx-bolts shall be used.

5.3. Design and Construction

5.3.1. Processes Guidelines

The subsystem components shall be designed to be installed unassisted or with articulated assistance according to assembly plant ergonomics requirements.

5.3.2. Parts Selection Guidelines

The suppliers will need to utilize existing fasteners and clamps.

5.3.3. Fasteners

Fasteners shall fulfill the following requirements:

- **Function requirement**
  The joint must have full satisfactory function with the minimum assembly torque, $M_v$min, after dynamic loading, and without the need to re-tighten the joint.

- **Inspection torque**
  The minimum inspection torque, $M_p$min, must be retained after assembly, and prior to dynamic loading of the joint.

- **Production process**
  The production process (e.g. positioning of spot weld, surface treatment, corrosion protection, etc.) must not negatively affect the function of the joint.

- **Tapped holes and weld nuts (if necessary)**
  Tapped holes, including weld nuts, must withstand a screw of property class 10.9. This requirement is valid for M10 joints and larger.

5.3.4. Ergonomics

a. Fabricating and assembling the design of the architecture and structure for subsystem require no red ergonomic stressors.
b. A score of yellow on any factor, or an unsatisfactory overall rating of yellow, or an existing plant ergonomic concern, shall require a detailed analysis of feasible product and/or process changes which will reduce the ergonomic stress to acceptable levels.
c. All parts shall be designed, so that the surface that interfaces with the operator is free from ridges, points, welding studs, burrs, or sharp edges.
d. The product shall not require the use of hammers, rubber mallets, etc to install parts.
e. The force required to be exerted by the palm or hand shall not exceed 8.2 kg. (Note: The force shall not exceed 4.5 kg to achieve green ergonomic value, which is the target for new designs.)
f. Product Engineers (Mechanical Engineering/Service) are responsible for measuring insertion forces of all manually installed parts.
g. The insertion force for snap fit parts designed or intended for manual installation shall not exceed 4.5 kg for a single digit or a finger/thumb pinch grasp. (Note: The force shall not exceed 1.0 kg to achieve green ergonomic value which is the target for new designs.)

The product design shall additionally provide visual, tactile or audible signals to the operator to support:
a. Identifying and selecting the correct part.
b. Assembling all the parts.
c. Correct function of the assembly.
d. Correct assembly orientation.

5.3.5. Quality Controls

The supplier should be informed that the items (relative distances, geometrical constraints, note on drawings, etc.) here listed are subjected to quality control KPC. Required quality controls have to be assured as listed in the attached file:

FDM_KPC.pdf

Each KPC/QH will be indicated on the drawing according to format at §2.3.